P1 $\psi_\pm(r_1, r_2) = B\left[\psi_a(r_1)\,\psi_b(r_2) \pm \psi_b(r_1)\,\psi_a(r_2)\right]$

(a) $$1 = |B|^2 \int d\vec{r}_1\, d\vec{r}_2 \left(\psi_a^*(r_1)\,\psi_b^*(r_2) \pm \psi_b^*(r_1)\,\psi_a^*(r_2)\right)\cdot$$

$$\left(\psi_a(r_1)\,\psi_b(r_2) \pm \psi_b(r_1)\,\psi_a(r_2)\right) = |B|^2 \Big\{ \underbrace{\int d\vec{r}_1\, |\psi_a(r_1)|^2}_{1} \underbrace{\int d\vec{r}_2 |\psi_b(r_2)|^2}_{1}$$

$$+ \underbrace{\int d\vec{r}_1\, |\psi_b(r_1)|^2}_{1} \underbrace{\int d\vec{r}_2\, |\psi_a(r_2)|^2}_{1} +$$

$$\pm \underbrace{\int dr_1\, \psi_a^*(r_1)\,\psi_b(r_1)}_{\phi} \underbrace{\int d\vec{r}_2\, \psi_b^*(r_2)\,\psi_b(r_1)}_{\phi}$$

$$\pm \underbrace{\int dr_1\, \psi_b^*(r_1)\,\psi_a(r_1)}_{\phi} \underbrace{\int d\vec{r}_2\, \psi_a^*(r_2)\,\psi_b(r_2)}_{\phi} \Big\} = |B|^2 \cdot 2$$

$$\Rightarrow B = \frac{1}{\sqrt{2}} \quad \text{for both signs.}$$

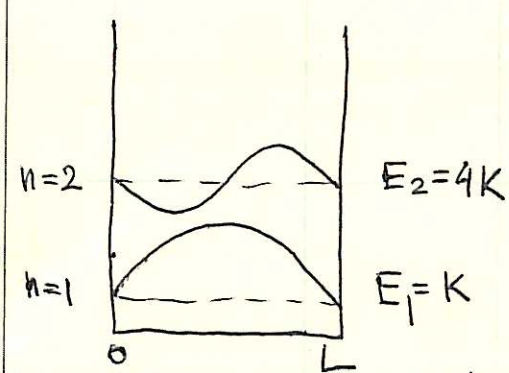
(b) If $\psi_a = \psi_b \Rightarrow \psi_+ = 2B\,\psi_a(r_1)\,\psi_a(r_2)$

$$(\psi_- = 0).$$

$$1 = \int d\vec{r}_1\, d\vec{r}_2\, |\psi(r_1, r_2)|^2 = (2B)^2 \int d\vec{r}_1\, |\psi_a(r_1)|^2 \int d\vec{r}_2\, |\psi_a(r_2)|^2 \Rightarrow$$

$$\Rightarrow 2B = 1 \Rightarrow B = 1/2.$$

P2



single particle states

$$\begin{cases} \psi_n(x) = \sqrt{\dfrac{2}{L}}\, \sin \dfrac{n\pi x}{L} \\ E_n = n^2 \dfrac{\pi^2 \hbar^2}{2mL^2} \equiv n^2 \cdot K \end{cases}$$

(a) $$\hat{H} = \frac{\hat{p}_1^2}{2m} + \frac{\hat{p}_2^2}{2m} = \left(-\frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2}\right)\frac{\hbar^2}{2m}$$

because $V(x) = \begin{cases} 0, & 0 < x < L \\ \infty, & \text{otherwise} \end{cases}$

$$\Psi_{ground}(x_1, x_2) = B\left[\psi_1(x_1)\,\psi_2(x_2) - \psi_2(x_1)\,\psi_1(x_2)\right]$$

(with $B = \frac{1}{\sqrt{2}}$ according to <u>P1a</u>)

Clearly $\hat{P}_{12}\,\Psi_{ground}(x_1, x_2) = -\Psi_{ground}(x_1, x_2)$.

Check the eigenvalue:

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x_1} + \frac{\partial^2}{\partial x_2^2}\right)\Psi_{ground}(x_1, x_2) = E_{ground}\,\Psi_{ground}(x_1, x_2)$$

$$\Psi_{ground} = \frac{1}{\sqrt{2}}\,\frac{2}{L}\left(\sin\frac{\pi x_1}{L}\sin\frac{2\pi x_2}{L} - \sin\frac{2\pi x_1}{L}\sin\frac{\pi x_2}{L}\right)$$

$$\frac{\partial^2}{\partial x_1^2}\Psi = -\frac{\sqrt{2}}{L}\left[\left(\frac{\pi}{L}\right)^2\sin\frac{\pi x_1}{L}\sin\frac{2\pi x_2}{L} - \left(\frac{2\pi}{L}\right)^2\sin\frac{2\pi x_1}{L}\sin\frac{\pi x_2}{L}\right]$$

$$\frac{\partial^2}{\partial x_2^2}\Psi = -\frac{\sqrt{2}}{L}\left[\left(\frac{2\pi}{L}\right)^2\sin\frac{\pi x_1}{L}\sin\frac{2\pi x_2}{L} - \left(\frac{\pi}{L}\right)^2\sin\frac{2\pi x_1}{L}\sin\frac{\pi x_2}{L}\right]$$

$$\Rightarrow \hat{H}\Psi = -\frac{\hbar^2}{2m}\left(-\frac{\sqrt{2}}{L}\right)\left\{\left[\left(\frac{\pi}{L}\right)^2 + \left(\frac{2\pi}{L}\right)^2\right]\sin\frac{\pi x_1}{L}\sin\frac{2\pi x_2}{L}\right.$$

$$\left. - \left[\left(\frac{2\pi}{L}\right)^2 + \left(\frac{\pi}{L}\right)^2\right]\sin\frac{2\pi x_1}{L}\sin\frac{\pi x_2}{L}\right\} =$$

$$= +\frac{\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2(1 + 2^2)\,\Psi(x_1, x_2)$$

$$\Rightarrow E_{ground} = \frac{\hbar^2}{2m}\frac{5\pi^2}{L^2} = 5\,K.$$

(b) The <u>first</u> excited state: antisymmetric combination of $n=1$ and $n=3$

$$\Psi_I(x_1, x_2) = B\left[\psi_1(x_1)\,\psi_3(x_2) - \psi_3(x_1)\,\psi_1(x_2)\right]$$

$$= \frac{1}{\sqrt{2}}\,\frac{2}{L}\left[\sin\frac{\pi x_1}{L}\sin\frac{3\pi x_2}{L} - \sin\frac{3\pi x_1}{L}\sin\frac{\pi x_2}{L}\right]$$

$$E_I = (1 + 3^2)K = 10\,K.$$

The <u>second</u> excited state : $n=2$ & $n=3$

$$\psi_{II}(x_1,x_2) = \frac{1}{\sqrt{2}}\left[\psi_2(x_1)\,\psi_3(x_2) - \psi_3(x_1)\,\psi_2(x_2)\right]$$

$$= \frac{\sqrt{2}}{L}\left(\sin\frac{2\pi x_1}{L}\sin\frac{3\pi x_2}{L} - \sin\frac{3\pi x_1}{L}\sin\frac{2\pi x_2}{L}\right)$$

with $E_{II} = (2^2+3^2)K = 13K.$

Yet the NEXT ONE (<u>third</u>) is made of $n=1$ and $n=4$: $\left[\psi_1(x_1)\psi_4(x_2) - \psi_4(x_1)\psi_1(x_2)\right]/\sqrt{2}$

with $E_{III} = (1+4^2)K = 17K.$

$\boxed{P3}$ = <u>P12.2</u> (Townsend).

$$\hat{H} = \underbrace{\frac{\hat{p}_1^2}{2m} + \frac{1}{2}m\omega^2 x_1^2}_{H_1} + \underbrace{\frac{\hat{p}_2^2}{2m} + \frac{1}{2}m\omega^2 x_2^2}_{H_2}$$

(a) $[\hat{H}_1, \hat{H}_2] = 0 \Rightarrow E = E_1 + E_2$

Two spin-½ particles $\Rightarrow \chi_{s=0} = \frac{1}{\sqrt{2}}\left(|+z,-z\rangle - |-z,+z\rangle\right)$

$$\chi_{s=1} = \begin{cases} |+z,+z\rangle & m=1 \\ \frac{1}{\sqrt{2}}\left(|+z,-z\rangle + |-z,+z\rangle\right) & \\ |-z,-z\rangle & \end{cases}$$

$\hat{P}\,\chi_{s=0} = -\chi_{s=0}$

but $\hat{P}\,\chi_{s=1} = \chi_{s=1}$

Using kets: $\chi_{s=0} \rightarrow |0,0\rangle = \frac{1}{\sqrt{2}}\left(|+z\rangle_1|-z\rangle_2 - |-z\rangle_1|+z\rangle_2\right)$

$\chi_{s=1} \Rightarrow |1,m\rangle \quad (m = +1, 0, -1)$

<u>Singlet</u> <u>Total spin – 0</u>

ground state $\psi(x_1, x_2; m_1, m_2) = \psi_0(x_1)\psi_0(x_2)\,|0,0\rangle$

$$E_{g.s.} = 2\cdot\left(\frac{\hbar\omega}{2}\right) = \hbar\omega$$

1st excited $\psi(x_1, m_1; x_2, m_2) = \frac{1}{\sqrt{2}}\left[\psi_0(x_1)\psi_1(x_2) + \psi_1(x_1)\psi_0(x_2)\right]\cdot$

$\cdot\,|0,0\rangle$

symmetric because $|0,0\rangle$ is <u>antisymmetric</u>

$$E_1 = \frac{\hbar\omega}{2} + \frac{3\hbar\omega}{2} = 2\hbar\omega$$

<u>Triplet</u> <u>Total spin – 1</u> $\hat{P}|1,m\rangle = +|1,m\rangle$

ground state $\Rightarrow$ spatial part must be <u>antisymmetric</u>

$$\psi_{g.s.}(x_1, m_1; x_2, m_2) = \frac{1}{\sqrt{2}}\left[\psi_0(x_1)\psi_1(x_2) - \psi_1(x_1)\psi_0(x_2)\right]\cdot$$

$\cdot\,|1,m\rangle$

$\rightarrow$ This state is 3-fold degenerate : $m = \pm 1, 0$.

$$E = 2\hbar\omega$$

(b) $V(|x_1 - x_2|) = \begin{cases} -V_0 & |x_1 - x_2| < a \\ 0 & \text{otherwise} \end{cases}$

$\Rightarrow$ the energy of <u>all</u> states will be lowered but differently: singlet ($S=0$) will come down more than triplet ($S=1$)

This is so because <u>symmetric</u> spatial part leads to a greater probability to find particles to each other in comparison with <u>antisymmetric</u> spatial wave functions.

$\boxed{P4} = \boxed{P12.8}$ Just check that $|\pm\rangle = \dfrac{|1\rangle \pm |2\rangle}{\sqrt{2 \pm 2c}}$, $c = \langle 1|2\rangle$

$\Rightarrow$ on that basis

$$\langle +|H|+\rangle = \frac{1}{2+2c} \langle 1| + \langle 2| \; H \; |1\rangle + |2\rangle = \frac{H_{11} + H_{12} + H_{21} + H_{22}}{2+2c} = \frac{H_{11} + H_{12}}{1+c}$$

$$\langle +|H|-\rangle = \frac{1}{\sqrt{2+2c}\sqrt{2-2c}} \langle 1| + \langle 2| \; \hat{H} \; |1\rangle - |2\rangle =$$

$$= \frac{H_{11} - H_{12} + H_{21} - H_{22}}{\sqrt{4-4c^2}} = 0 \quad \text{because} \quad \begin{matrix} H_{11} = H_{22} \\ H_{12} = H_{21} \end{matrix}$$

$\Rightarrow \langle -|H|+\rangle = 0$ as well.

$$\langle -|H|-\rangle = \frac{1}{2-2c} \langle 1| - \langle 2| \; H \; |1\rangle - |2\rangle = \frac{H_{11} - H_{12} - H_{21} + H_{22}}{2-2c} = \frac{H_{11} - H_{12}}{1-c}$$

Thus $\hat{H} \rightarrow \begin{pmatrix} \langle +|H|+\rangle & \langle +|H|-\rangle \\ \langle -|H|+\rangle & \langle -|H|-\rangle \end{pmatrix} =$

$$= \begin{pmatrix} \dfrac{H_{11} + H_{12}}{1+c} & 0 \\ 0 & \dfrac{H_{11} - H_{12}}{1-c} \end{pmatrix}$$

which is just equation (12.72)